Onda stazionaria

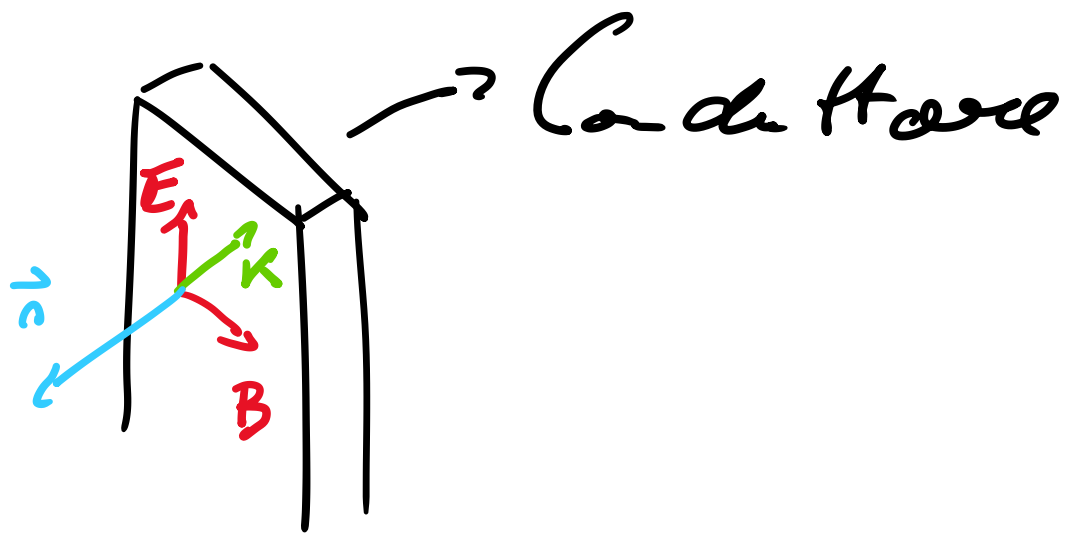
2/14/21 3:52 PM

$[D_n] = \sigma$

$[B_n] = 0$

$[E_t] = 0$

$[H_t] = J_s$



$E_{int} = 0$ ONDA PIANA

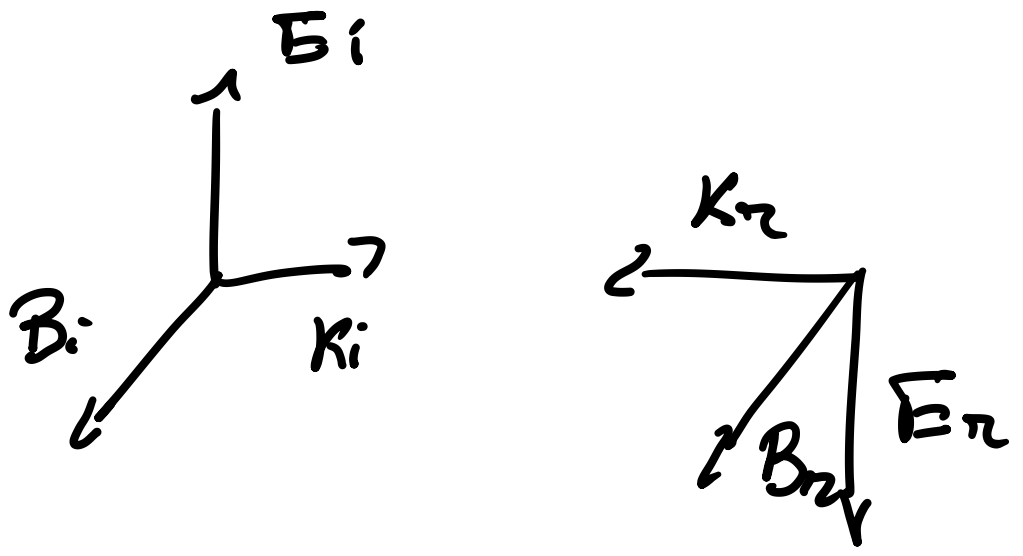
$B = \frac{1}{c} \bar{E} \times \bar{n}$

$E_{int} = 0$

Non c'è componente che si propaga

$\nabla \times E = -\frac{\partial B}{\partial t} \Rightarrow B_{int} = 0$ (costante)

Per le cond al contorno



Sulla Σ

$\bar{E} = 0$ $\qquad E_r = -E_i$

$\bar{B} = 2B_i$ $\qquad B_r = B_i$

$E_i = E_0 \operatorname{sen}(\omega t - kz)$ $\qquad E_r = -E_0 \operatorname{sen}(kz + \omega t)$

$$\operatorname{sen}(\alpha) - \operatorname{sen}(\beta) = 2\cos\left(\frac{\alpha+\beta}{2}\right)\operatorname{sen}\left(\frac{\alpha-\beta}{2}\right)$$

$$\bar{E} = 2E_0 \cos\left(\frac{\omega t - \cancel{kz} + \omega t + \cancel{kz}}{2}\right) \operatorname{sen}\left(\frac{\cancel{\omega t} - kz - \cancel{\omega t} - kz}{2}\right)$$

$$= \underbrace{-2E_0 \cos(\omega t)\operatorname{sen}(kz)}_{\text{ONDA STAZIONARIA}}$$